ITEM1

The acidity of water can be determined by neutralisation using an acid–base reactio n. The volume of hydrochloric acid required to neutralise a fixed volume of sodium hy droxide can be used to determine the concentration of the acid in the water.

Residents in Kasese who obtain water from a source near a textile factory are concer ned that acidic waste from the factory may have affected the water. A technician advi sed them to investigate the acidity of the water by determining whether the volume of the water sample is greater than 25cm3 of the alkali used.

Local leaders requested your school laboratory team to help with the investigation. Yo ur chemistry teacher collected a water sample from the source and provided the follo wing materials:

BA1: the water from the source near the factory.

BA2: Sodium hydroxide solution.

Relevant apparatus available in a chemistry laboratory.

TASK

As a chemistry learner, design and perform an experiment to determine the concentrat ion levels of BA1 and advise the local leaders on the acidity of the water.

As a chemistry learner:

EXPECTED RESPONSES

(a) Design and perform the experiment

AIM:

An investigation to determine the volume BA1 that neutralizes a fixed volume of BA2 to see if the water is highly polluted and acidic by measuring volume of BA2 that reac ts with a fixed volume of BA1

HYPOTHESIS:

The volume of BA2 that reacts with a fixed volume of BA1 is greater than 25cm3 dete

rmined by measuring volume of BA2 that reacts with a fixed volume of BA1.

VARIABLES

Independent:
Volume of BA2 pipetted.

Dependent:
Volume of BA1 required to react completely with BA2.

Controlled:

- Volume of BA2 used kept constant at 25.0cm^3

PROCEDURE

1. A burette was washed and rinsed with water and then filled with BA1.
2. The initial burette reading was recorded.
3. A pipette was used to transfer **25.0 cm³ of BA2** into a conical flask.
4. BA2 was titrated with BA1 using phenolphthalein indicator while swirling the fl ask until the solution turned to colourless.
5. The final burette reading was recorded.
6. The titre was calculated from the difference between the final and initial readin gs.
7. The titration was repeated until **consistent values** were obtained.

RISKS AND MITIGATIONS

Ris k	Mitigatio n

(b) Experimental data

Volume of pipette used $= 25.0\text{cm}^3$.

Experiment number	*1*	*2*	*3*
Final burette reading (cm^3)	*25.50*	*41.10*	*46.60*

Initial burette reading (cm^3)	0.00	15.50	21.10
Volume of BA1used (cm^3)	25.50	25.60	25.50

(c) Analysis, interpretation,

values for calculating the average volume of BA1 used 25.50, 25.50cm^3 .

$$\text{average volume} = \frac{25.50 + 25.50}{2} .$$

$$= 25.50 cm^3 .$$

conclusion and recommendation

Conclusion:
The volume of BA2 that reacts with a fixed volume of BA1 is greater than 25cm3 and therefore highly polluted.

Recommendation:
The local leaders should be **discouraged to use the water it is not safe.**

OR

ITEM2

The acidity of water can be investigated using a neutralization reaction in which sodi um hydroxide reacts with hydrochloric acid. The masses of the two reactants that rea ct completely are directly proportional according to the relation.

40 g of sodium hydroxide reacts with 36.5 g of hydrochloric acid.

Residents in Kasese who obtain water from a source near a textile factory using hydr ochloric acid, fear that acidic waste may have contaminated the water. A technician a dvised them to determine whether the mass of hydrochloric acid in the water is at lea st 30 g per 100 cm^3 of water used by reacting the water sample with sodium hydroxi de solution.

Local leaders requested your school laboratory team to help with the investigation. Yo ur chemistry teacher provided the following materials:

- BA1: Water sample obtained from the source near the factory, containing h ydrochloric acid.
- BA2: Sodium hydroxide solution containing 0.4 g of sodium hydroxide per 100 cm^3.
- Relevant apparatus available in a chemistry laboratory.

TASK

As a chemistry learner, design and perform an experiment to determine the mass of h ydrogen chloride in BA1 and advise the local leaders on whether the water contains at least 3 g of hydrochloric acid per 100 cm^3.

Note: The stated threshold of 3 g per 100 cm^3 is an experimental criterion for this sce nario, not a safe drinking-water limit.

EXPECTED RESPONSES

(a) Design and perform the experiment

AIM:
An investigation to determine the mass of HCl in BA1 to see if the water is highly poll uted and acidic by measuring volume of BA2 that reacts with a fixed volumeof BA1

HYPOTHESIS:
The amountof HCl in BA1 is **more** than **3g in 100cm^3 determined** by measuring volu me of BA2 that reacts with a fixed volume of BA1.

VARIABLES

Independent:
Volume of BA2 pipetted.

Dependent:
Volume of BA1 required to react completely with BA2.

Controlled:

- Volume of BA2 used kept constant at 25.0cm^3

PROCEDURE

8. A burette was washed and filled with BA1.

9. The initial burette reading was recorded.

10. A pipette was used to transfer **25.0 cm^3 of BA2** into a conical flask.

11. BA1 was titrated with BA2 using phenolphthalein indicator while swirling the fl ask until the solution turned to pink.

12. The final burette reading was recorded.

13. The titre was calculated from the difference between the final and initial readin gs.

14. The titration was repeated until **consistent values** were obtained.

RISKS AND MITIGATIONS

Ris k	Mitigatio n

(b) Experimental data

$Volume\ of\ pipette\ used\ =\ 25.0cm^3$.

Experiment number	1	2	3
Final burette reading (cm^3)	25.50	41.10	46.60
Initial burette reading (cm^3)	0.00	15.50	21.10
Volume of BA1 used (cm^3)	25.50	25.60	25.50

(c) Analysis, interpretation,

values for calculating the average volume of BA1 used 25.50, $25.50cm^3$.

$$\text{average volume } = \frac{25.50 + 25.50}{2} .$$

$$= 25.50cm^3 .$$

$100cm^3$ of solution contain $0.4g$ of $NaOH$

$25.5cm^3$ of solution contain $\frac{0.4 \times 25.5}{100}$ g of $NaOH$

$= 0.102g$ of $NaOH$

from the ratio :

$40g$ of oxidising agent reacts with $36.5g$ of HCl

$0.0496g$ of oxidising agent react with $(0.102X36.5/40)\ g$ of HCl .

$= 0.093075g$

$25.0cm^3$ of solution contains $0.093075g$ of HCl

$100cm^3$ of solution contain $\frac{0.093075 \times 100}{25.0}$ g of HCl

$=\mathbf{0.3723\ g}$

$\text{amount of } NaOH \text{ in } 100cm3 = \ 0.3723g$

conclusion and recommendation

Conclusion:
The water sample from near the factory contains **less than 3g of NaOH** and therefore not much polluted.

Recommendation:
The local leaders should be **encouraged to use the water it is safe.**

ITEM3

The energy changes accompanying chemical reactions determine the heat released o r absorbed during industrial chemical processes.

At a secondary school in Mukono, S.2 learners preparing sodium chloride in the chem istry laboratory noticed and got surprised that the reaction vessels became hotter wit hout external heating. Their chemistry teacher, Mr. Kato, contacted the laboratory tech nician to help the learners investigate this and determine the amount of heat if its mo re than 5 kJ of heat for 50cm3 of sodium hydroxide used to ensure safety.

$HCl\,(aq) + NaOH(aq)$ $NaCl(aq) + H_2O(l) + heat$

The lab technician has guided you to ddesign an investigation by measuring tempera ture of the mixture when BA1 was added to BA2 in uniform volume intervals.
Hypothesis:
You are provided with:

- **BA1** which is hydrochloric acid
- **BA2** which is sodium hydroxide

TASK:
Copy the lab technician to design and carry out an experiment that would help the lea rners.
Aim:

To determine the amount of heat given out during a reaction of sodium hydroxide an d hydrochloric acid by measuring temperature of the mixture when varying volumes o f BA1 was added to a fixed volume of BA2 in uniform intervals.

Hypothesis:

The reaction between sodium hydroxide and hydrochloric acid produces more than 5 kJ of heat for 50cm^3 of sodium hydroxide determined by measuring temperature of th e mixture when varying volumes of BA1 was added to a fixed volume of BA2 in unifor m intervals.

Variables:

- The dependent variable: Temperature of the mixture.
- The independent variable: Volume of **BA1.**
- The controlled variable: Volume of **BA2** which is kept constant at 50.0cm^3.

Risks and mitigation

- Skin burns and irritations due to hydrochloric cid which is corrosive pouring on the skin.

  Mitigated by wearing a laboratory coat, gloves and closed shoes to prevent co ntact of the chemical with the skin.
- Digestive disorder due to swallowing of the base or acid during pipetting alteri ng pH in the digestive system.

  Mitigation: Use a pipette sucker or filler toavoid swallowing the chemical

Procedures

(a) 30cm3 of **BA2** was measured using a measuring cylinder and put into a clean plastic beaker

(b) The initial temperature of the solution **BA2** was measured using a thermomete r

(c) The initial temperature of **BA1** is also noted and recorded and then filled into a burette and adjusted to the zero mark.

(d) **BA1** was added to **BA2** in the beaker at uniform intervals of 10cm3 each time stirring and noting the highest temperature of the mixture for seven readings u pto 50cm3.

(e) The results were tabulated.

Data recording and presentation of data

Initial temperature of **BA1** = 27.5 0C

Initial temperature of **BA2** = 27.5 0C

Volume of BA2= 30.0 cm^3

Volume of **BA1**	0.00	10.00	20.00	30.00	40.00	50.00
Maximum temperature of mixture (0C)	27.5	33.0	37.0	34.5	33.0	32.5

Data analysis and interpretation

Heat change for a reaction between hydrochloric acid and sodium hydroxide

$$\text{Highest temperature } = 36.8^0\text{C}$$

$$\text{mass of the solution} = (24 + 30) \times 1 = 54\text{g}$$

$$\text{Heat produced} = \text{mass} \times \text{specific heat capacity} \times \text{temperature change}$$

$$= 54 \times 4.2 \times (36.8^0\text{C} - 27.5^0\text{C}) = -2259.9 \text{ J}$$

$$30 \text{ cm3produces} - 2259.9 \ J$$

$$50 \text{ cm3produces} - 2259.9\text{x } 50/30 \ J$$

$$= -3766.5 \ J$$

$$\text{Heat produced for 50cm3 of BA2} = -3766.5 \ J$$

Conclusion and recommendation

A reaction between sodium hydroxide and hydrochloric acid produces less than 5 kJ of heat for 50cm^3 of sodium hydroxide determined. So, it safe for use by the s2 learne rs.

A graph of volume of BA1 against maximum temperature of the mixture

Vertical scale: 1cm represents 0.5^0C

Horizontal scale: 1cm represents $5cm^3$

